

Exercise 1.

1st SECTION (XXX): $\binom{26}{2} \times 10 \times 3!$

2nd SECTION (XXXX): 10^4

= 1st Section $\times$ 2nd Section

= $(\frac{26!}{2!24!} \times 10 \times 3!)(10^4)$

= $(325 \times 10 \times 6)(10,000)$

= $(19,500)(10,000)$

= 195,000,000

Exercise 2.

BLAZER: color 1, color 2, color 3

PANTS: color 1, color 2, color 3

SHIRTS: 2 white, 1 pink, 1 purple, 1 black

- $3/5$ - chance to choose the right colored shirt (2 white + 1 black)
- 1 - whichever is chosen first blazer or pants the color doesn't matter
- $1/3$ - chance to pick the same color as either the blazer or pants that is already chosen

= $\frac{3}{5} \times 1 \times \frac{1}{3}$

= $\frac{3}{15}$

= $\frac{1}{5}$

Bill has a $1/5$ chance that he makes a good impression.

OR

Total number of combinations: $3 \times 3 \times 5 = 45$

Number of good impression outfits: $3 \times 3 = 9$

$$\frac{\text{number of good impression outfits}}{\text{total number of combinations}} = \frac{9}{45} = \frac{1}{5}$$

Bill has a $1/5$ chance that he makes a good impression.

Exercise 3.

(a) $|\Omega| = 4^{12} = 16,777,216$

(b) We want to count the number of strings with 3 of each character (1, 2, 3, and 4)

$$\frac{12!}{3!3!3!3!} = \frac{479,001,600}{1,296} = 369,600$$

$$\begin{aligned}
 (c) &= \frac{\text{number of strings with 3 of each character}}{|\Omega|} \\
 &= \frac{369,600}{16,777,216} \\
 &\approx 0.0220298767
 \end{aligned}$$

Exercise 4.

$$= 5!4!2!1!4! = 138,240$$

Exercise 5.

1st card is also the same suit: $\frac{13}{52} \times \frac{12}{51} \times \frac{11}{50}$

1st card isn't the same suit: $\frac{39}{52} \times \frac{13}{51} \times \frac{12}{50}$

$$\frac{13 \times 12 \times 11 + 39 \times 13 \times 12}{52 \times 51 \times 50} = \frac{1716 + 6084}{132600} = \frac{1}{17}$$

Exercise 6.

$$1. 9! = 362,880$$

$$2. 5 \text{ girls and } 4 \text{ boys}$$

if alternating girls must be first

$$\begin{aligned}
 &(g_1, b_1, g_2, b_2, g_3, b_3, g_4, b_4, g_5) \\
 &= 5 \times 4 \times 4 \times 3 \times 3 \times 2 \times 2 \times 1 \times 1
 \end{aligned}$$

or

$$\begin{aligned}
 &= 5!4! \\
 &= 2,880
 \end{aligned}$$

$$3. 8!2! = 80,640$$

$$4. 6!4! = 17,280$$

$$5. 5!5! = 14,400$$

$$6. 2!4!5! = 5,760$$

$$7. 10! = 3,628,800$$

$$8. \text{ Either boys or girls can be first}$$

$$(g_1, b_1, g_2, b_2, g_3, b_3, g_4, b_4, g_5, b_5)$$

or

$$\begin{aligned}
 &(b_1, g_1, b_2, g_2, b_3, g_3, b_4, g_4, b_5, g_5) \\
 &= 2 \times 5!5!
 \end{aligned}$$

$$= 28,800$$

Exercise 7.

(a) PLEXIS = 1 Only one combination can make PLEXIS

$$\frac{1}{6!} = \frac{1}{720}$$

$$(b) = \frac{5!}{2!2!} = \frac{120}{2 \times 2}$$

$$\frac{120}{4}$$

$$= 30$$

There is only 1 instance of MIAMI

$$= \frac{\text{number of instances of MIAMI}}{\text{total number of arrangements}}$$

$$= \frac{1}{30}$$

Exercise 8.

(a) BOOLAHUBBOO = 11 letters

$$B = 3$$

$$O = 4$$

$$L = 1$$

$$A = 1$$

$$H = 1$$

$$U = 1$$

$$\frac{11!}{3!4!1!1!1!1!} = \frac{39916800}{144} = 277,200$$

(b) The 5 remaining letters are distinct

$$5!1! = 120$$

$$(c) = \frac{9!}{4!1!1!1!1!1!}$$

$$= \frac{362,880}{24} = 15,120$$

(d) 6 vowels: O, O, A, O, O, U

$$= \frac{4!6!}{4!}$$

$$= \frac{17,280}{24}$$

$$= 720$$

Exercise 9.

$$\begin{aligned} &= \frac{10!}{3!2!5!} \\ &= \frac{3,628,800}{1,440} \\ &= 2,520 \end{aligned}$$

Exercise 10.

52! combinations to shuffle a deck

(a) All suits in order: $= 13!13!13!13!4!$

(b) $= 40!13!$

Exercise 11.

$$\binom{40}{5} = \frac{40!}{5!(40-5)!} = \frac{40!}{5!35!} = 658,008$$

Exercise 12.

$$\binom{21}{4} = \frac{21!}{4!(21-4)!} = \frac{21!}{4!17!} = 5,985$$